Project Progress Tracker Lab Report

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1. Project Description

Problem Statement

The ability to keep track of the progress for different projects at the same time, including milestones and collaborators.

Target Users

The target users are anyone currently working on a project. More specifically, I would associate this with graduate students in the IT Design Studio class as a reference.

Key Features

Create, view, update, and delete project records.

System Architecture

- Front End: Angular
- Back End: NodeJS and ExpressJS
- Database: MongoDB

2. Project Design

Database Schema

Projects Collection:

- projectName: String
- description: String
- startDate: Date
- status: String
- teamMember: String
- progressNote: String

User Stories and Acceptance Criteria

Create Project Record:

- Story: The ability to add a new project to the project list in the database without affecting other projects.
- Acceptance Criteria: The criteria would be the project appearing in the project list with the correct information, none of the other projects are affected, and the record remains there after the user has left the project.

View Project Records:

- Story: The ability to view past records made at any time, including all six fields.
- Acceptance Criteria: The project record is retrievable from the database with all the information in the six fields properly maintained.

Update Project Record:

- Story: The ability to edit a prior entry without removing the entry from the database.
- Acceptance Criteria: Any of the six fields can be changed and those changes overwrite the saved entry in the database.

Delete Project Record:

- Story: To remove an entry from the database without affecting the other entries.
- Acceptance Criteria: When the user can completely remove an entry from the database without affecting other entries and when the user reuses the application, that entry is still deleted from the database.

3. Project Evaluation

Results

Design

The user's needs were met. The schema encompasses all the necessary fields to completely describe a project for record-keeping. The capabilities programmed in the application allow users complete control of the information entered.

Implementation

The application was developed using the best practices. The synergy of a Angular as the front-end with NodeJS and ExpressJS as the back end is high and known due to the common MEAN

stack that they are utilized in. MongoDB is a reliable open-source database that is still being maintained.

Functionality

Application successfully meets all user requirements regarding the ability to create, read, update, and delete any entries.

Evaluation

Success

The project was successful in meeting it's objectives.

Challenges

A challenge that occurred was properly installing MongoDB and ensuring that the information from ExpressJS was going to the right database.

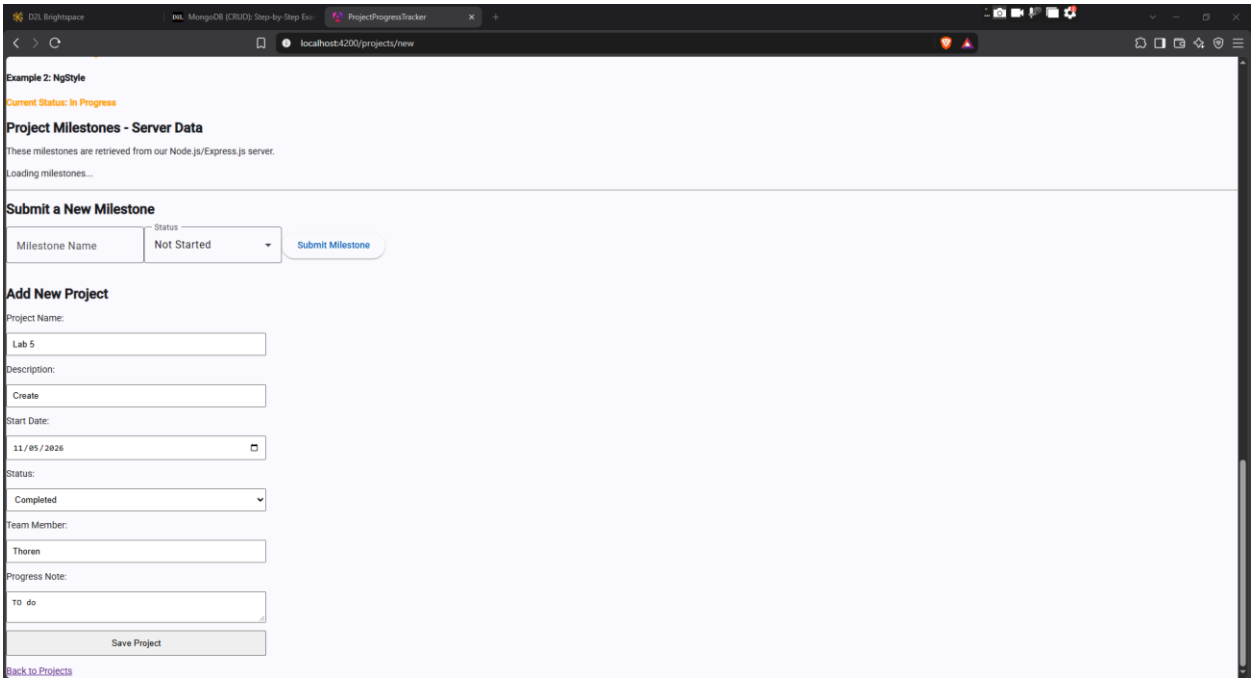
Improvements

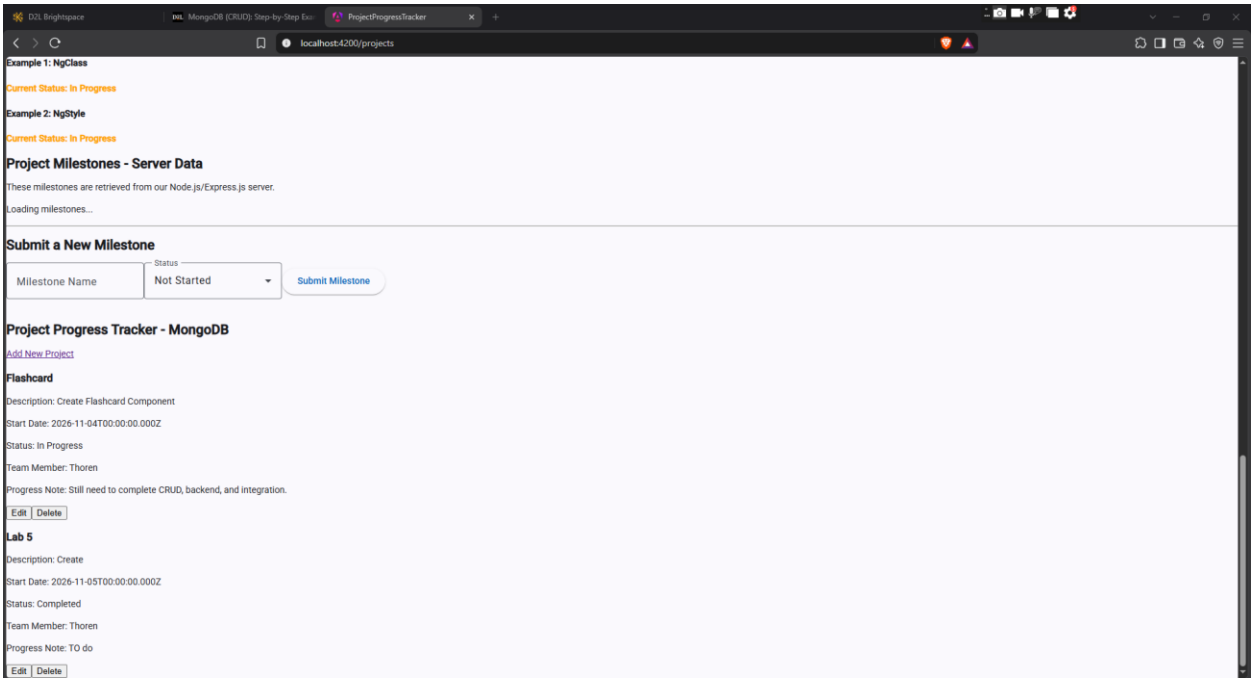
An improvement is adding the ability to hide the project details of entries that the user is not focusing on.

4. Screenshots and Explanation

Include the following five screenshots demonstrating the successful execution of your application.

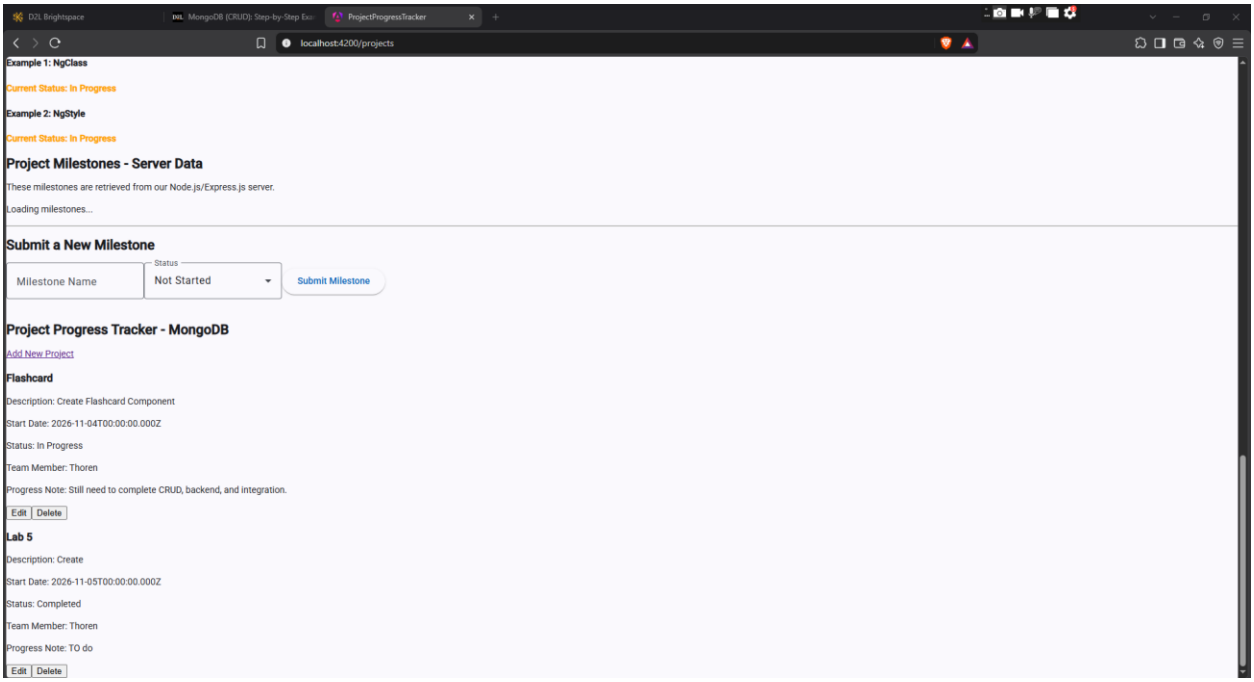
Screenshot 1: Create (POST)





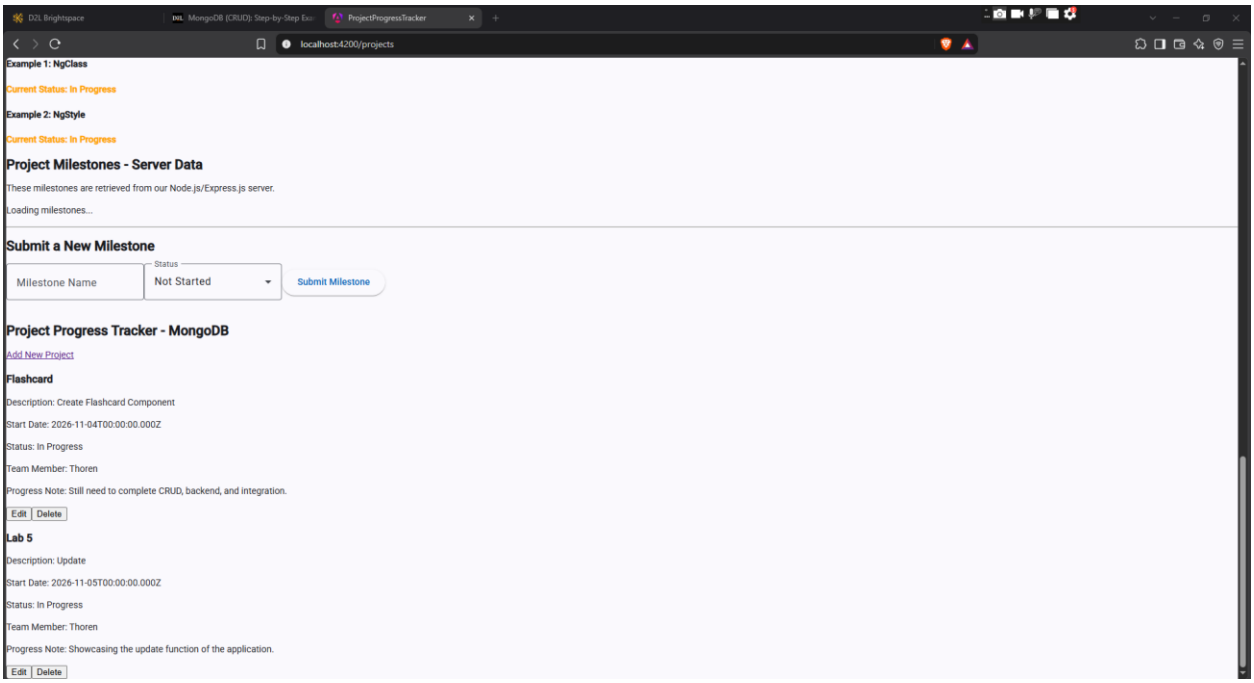
The angular form accepts the user inputs through form controls submitted by the save button. Express.JS creates a POST request based on the user inputs obtained through CORS and submits the POST request and the information is saved to MongoDB through a link provided in the application code by project.ID.

Screenshot 2: Read (GET)



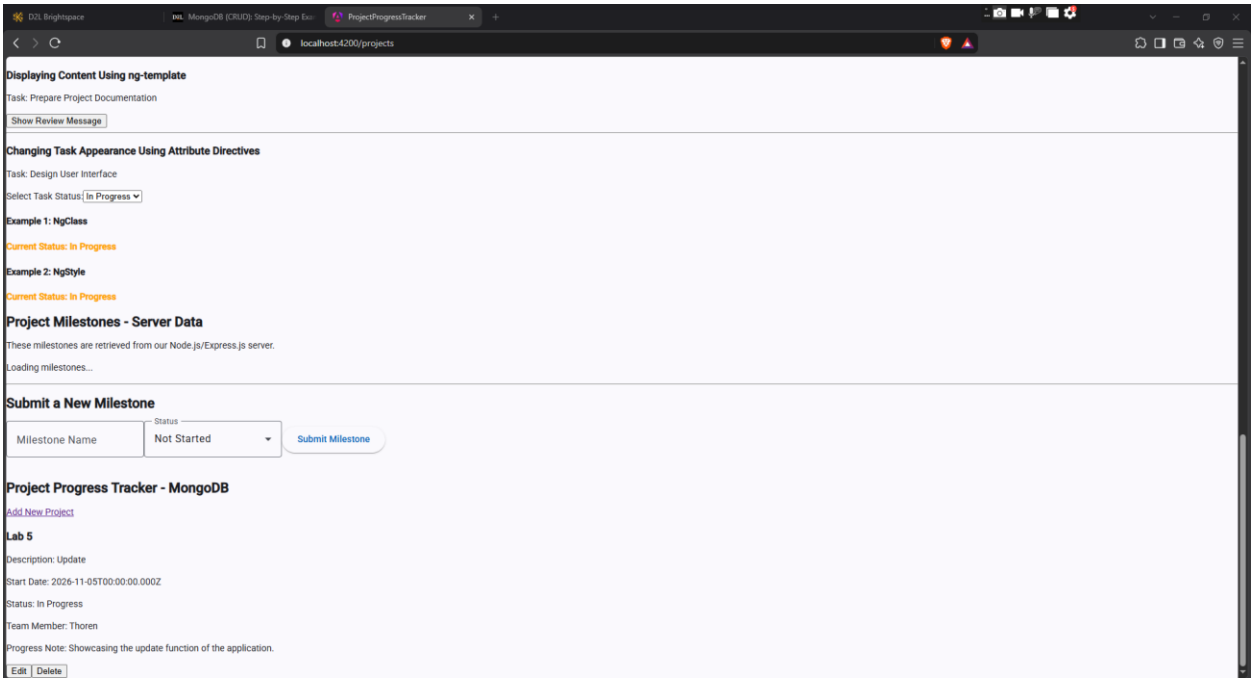
On the initialization of the application, a function called getProjects runs to retrieve the projects from an HTTP link to MongoDB. This link utilizes Express.JS to send a GET request to the link for all projects through their project.IDs.

Screenshot 3: Update (PUT)



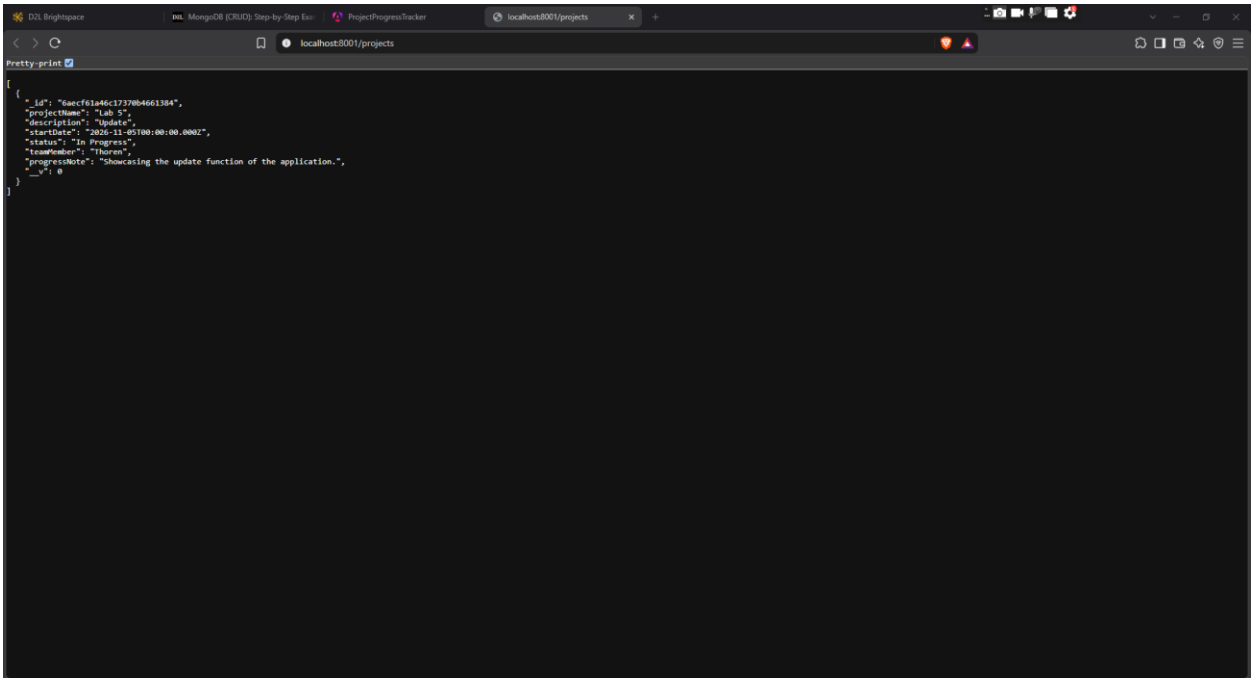
User clicks the Edit button which opens the router links edit form. Once the update button is pressed to submit the changes, Angular submits the JSON information to Express and Express creates a PUT request that's sent to MongoDB.

Screenshot 4: Delete (DELETE)



When the user presses the delete button, the application runs the onDelete function referencing the project ID of the record. This function retrieves the project ID and JSON information and sends it to Express for Express to send a request through HTTP to MongoDB to delete the project from the database. When the deletion is complete the server returns a result based on the success of the request.

Screenshot 5: MongoDB Records (Express GET Endpoint)



On the initialization of the application, Express will connect to MongoDB through the 27017 port. Then Express sends a GET request for the project JSON information through the HTTP client and displays the results on the running server on port 8001.